NLP Project Progress Update

TempoRAG-KG

Temporal Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering

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From Wikipedia QA to SEC 10-K filings

Dataset pivot: anchoring problem → year-stamped corpus

	Proposal round	Progress round
Dataset	HotpotQA + MuSiQue (Wikipedia multi-hop)	25 × 10-K filings (5 tech mega-caps × 5 fiscal years)
Core limitation we hit	Facts lack explicit year anchors; Wikipedia spans decades with no canonical "as of" date	Every 10-K is filed for a specific fiscal year — year is metadata, not inference
Why it matters for us	Can't evaluate temporal filtering when ground-truth time itself is ambiguous	Temporal correctness is testable — filter right year, get right fact

Research questions & experimental design

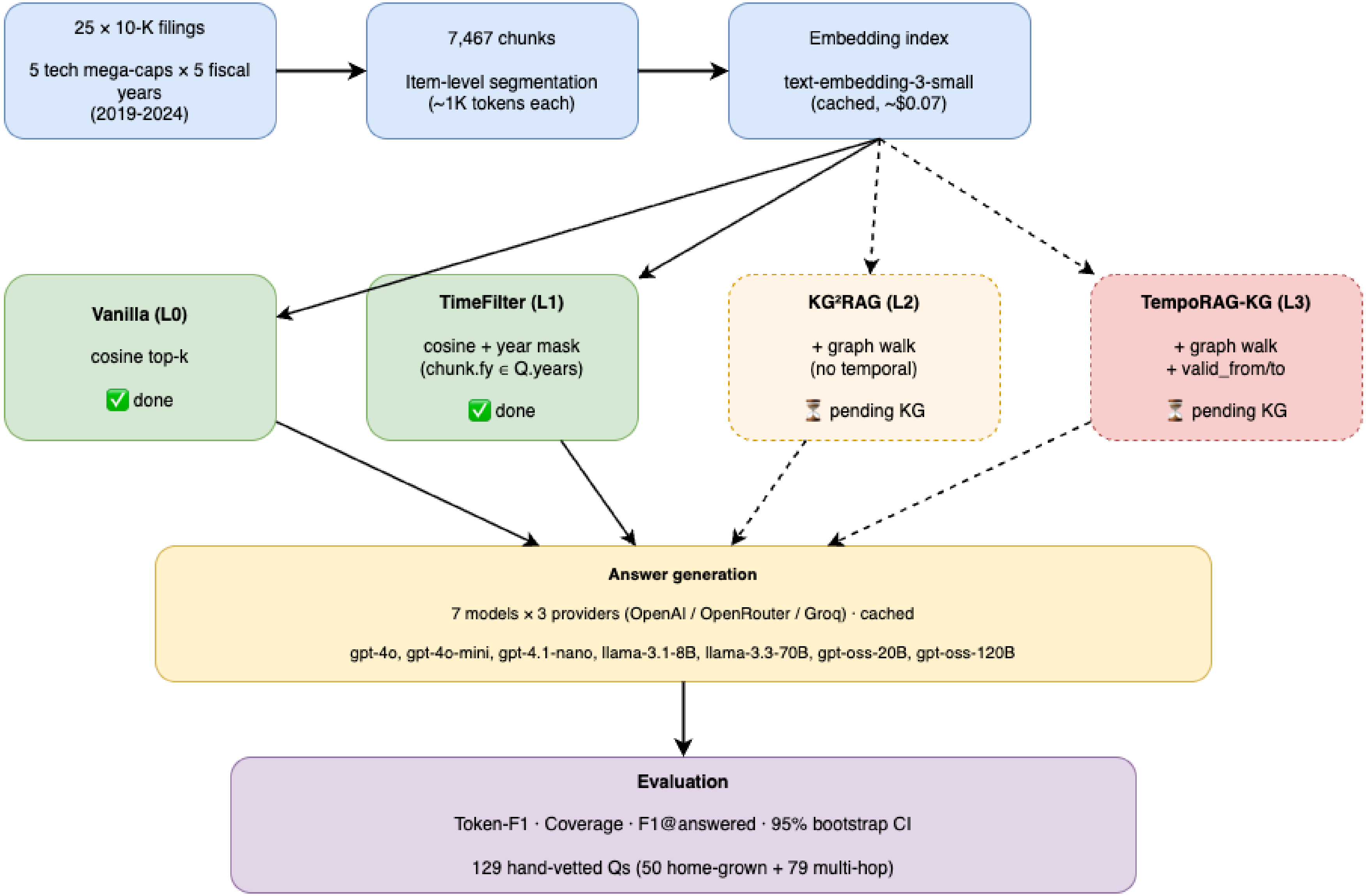
#	Question
RQ1	Where does KG²RAG fail on temporal multi-hop QA?
RQ2	Does TempoRAG-KG lift F1 over KG²RAG?
RQ3	How accurate is temporal extraction?
RQ4 ★	**Can small-LM + temporal-KG ≈ large-LM alone?**

RQ4 hypothesis: *Temporal grounding closes the small-vs-large capability gap on temporal QA.*

Test: 2 × 2 factorial · 7 models · 129 Qs

	No KG	**+ KG**
No Temporal	Vanilla	KG²RAG
+ Temporal	TimeFilter	TempoRAG-KG

Pipeline


25 × 10-K filings
5 tech mega-caps × 5 fiscal years
(2019-2024)
7,467 chunks
Item-level segmentation
(~1K tokens each)
Embedding index
text-embedding-3-small
(cached, ~$0.07)
Vanilla (L0)
cosine top-k
✅ done
TimeFilter (L1)
cosine + year mask
(chunk.fy ∈ Q.years)
✅ done
KG²RAG (L2)
+ graph walk
(no temporal)
⏳ pending KG
TempoRAG-KG (L3)
+ graph walk
+ valid_from/to
⏳ pending KG
Answer generation
7 models × 3 providers (OpenAI / OpenRouter / Groq) · cached
gpt-4o, gpt-4o-mini, gpt-4.1-nano, llama-3.1-8B, llama-3.3-70B, gpt-oss-20B, gpt-oss-120B
Evaluation
Token-F1 · Coverage · F1@answered · 95% bootstrap CI
129 hand-vetted Qs (50 home-grown + 79 multi-hop)

Temporal filtering lifts F1 on all 7 models (+5.6% to +24.7%)

L0 Vanilla vs L1 TimeFilter · 129 Qs · 95% bootstrap CIs available

Model	Vanilla F1	+TimeFilter	Δ F1 %	Van cov	TF cov	Δ cov	F1@ans V→T
gpt-4.1-nano	0.229	0.258	**+12.7%**	63.6%	70.5%	**+7.0pp**	*0.360 → 0.366*
gpt-4o-mini	0.230	0.248	**+7.8%**	58.1%	62.8%	**+4.7pp**	*0.396 → 0.395*
gpt-4o	0.183	0.221	**+20.9%**	47.6%	56.6%	**+9.0pp**	*0.385 → 0.391*
llama-70b	0.145	0.179	**+23.7%**	43.4%	53.5%	**+10.1pp**	*0.333 → 0.335*
llama-8b	0.145	0.180	**+24.7%**	48.1%	57.4%	**+9.3pp**	*0.301 → 0.314*
gpt-oss-120b	0.131	0.138	**+5.6%**	35.7%	38.8%	**+3.1pp**	*0.366 → 0.356*
gpt-oss-20b	0.120	0.145	**+21.0%**	37.2%	42.6%	**+5.4pp**	*0.320 → 0.336*

Universal lift — 7/7 models improved (+5.6% to +24.7% F1).

Coverage drives F1 — Δ cov +3 to +10pp; F1@answered stays flat ≈ 0.35. Bottleneck is retrieval, not answer quality.

Small-model benefit — llama-8b +24.7%, gpt-oss-20b +21.0% vs gpt-4o-mini +7.8%. Early signal for RQ4 capability parity.

Where TimeFilter hits its ceiling

gpt-4.1-nano slicing · motivates why we need KG next

Hop count	Vanilla F1	TimeFilter F1	Δ	n
hop = 1	0.337	0.206	-0.131	10
hop = 2	0.234	0.281	+0.048	104
hop = 3	0.123	0.131	+0.007	15

Scope	Vanilla F1	TimeFilter F1	Δ	n
inter_year	0.220	0.276	+0.055	53
cross_company	0.264	0.315	+0.051	24
intra	0.216	0.216	-0.001	45
fiscal_vs_calendar	0.301	0.222	-0.079	5
forward_looking	0.138	0.157	+0.019	2

TimeFilter shines on year-sensitive questions — inter_year +0.055, cross_company +0.051. Exactly where a year mask should help.

No lift on intra (same-year) questions — Δ ≈ 0. Year filter is a no-op when the question and all candidate chunks share one year.

hop = 3 barely moves — Δ = +0.007 (n=15). Multi-hop reasoning needs entity cross-linking, which metadata alone cannot provide.

→ **This motivates KG²RAG and TempoRAG-KG** — graph walks add the cross-chunk relations TimeFilter is blind to.

What's next — KG extraction and the L2 / L3 lanes

KG extraction (running)

Stage	Detail
Corpus	7,467 chunks from 25 × 10-K filings
Extractor	gpt-4.1-nano, temporal-aware prompt
Schema	(subject, predicate, object, valid_from, valid_to)
Progress	Running (live + cached), mid-sweep
Budget	Well under $20 cap

Remaining work

1. **Finish extraction** — complete full 7,467-chunk pass.
2. **Filter pass** — drop noisy triples (e.g. object="true", comparative artifacts).
3. **Build KG²RAG retriever** — seed chunk → graph walk → expanded context.
4. **Build TempoRAG-KG retriever** — KG²RAG + hard mask on valid_from / valid_to.
5. **Run L2 + L3 sweeps** — same 7 models × 129 Qs.
6. **Interaction analysis** — hop × scope × condition, test whether L3 > L1 + L2.

Takeaways

From this sprint · L0 + L1 baselines on 10-K temporal QA

1. **TimeFilter baseline proven** — +5.6% to +24.7% F1 across 7 models; lift is universal, not model-specific.
2. **Retrieval, not answer quality, is the bottleneck** — F1@answered stays flat around 0.35 while coverage moves +3 to +10pp. Any further lift must come from better retrieval.
3. **Small models benefit more than large** — llama-8b +24.7% vs gpt-4o-mini +7.8%. Directional support for the RQ4 capability-parity story.
4. **Multi-hop is next** — TimeFilter cannot lift hop = 3 (+0.007). KG²RAG and TempoRAG-KG will target exactly this gap.